`legged_mani`: B2–Z1 Whole-Body MPPI
Mathematical Formulation and Implementation Analysis

Notes based on analysis of the `legged_mani` codebase

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Contents

1 Scope and Executive Summary

This document analyzes the `locomani` task in `legged_mani`, a whole-body Model Predictive Path Integral (MPPI) controller for a Unitree B2 quadruped equipped with a reduced four-degree-of-freedom Unitree Z1 arm. The analysis is based primarily on the following files:

- MPPI sampling and rollout: `control/controllers/base_controller.py`,
- locomani controller: `control/controllers/mppi_locomani.py`,
- task and solver parameters: `utils/tasks.py` and `configs/mppi_locomani.yml`,
- robot and rollout models: `models/b2_z1_4dof.xml` and `models/b2_z1_base.xml`,
- closed-loop execution: `interface/simulator.py`.

The main architectural decisions are:

1. The observation is the complete 45-dimensional MuJoCo position and velocity state.
2. The optimized input is a vector of 16 desired joint positions, not joint torques or contact forces.
3. Full multibody dynamics and contact are evaluated through parallel MuJoCo rollouts instead of an explicit reduced-order dynamics model.
4. The leg nominal trajectory is supplied by a periodic gait file, while the arm nominal posture is generated by damped least-squares inverse kinematics (IK) for the current end-effector target.
5. Each rollout is scored by body and joint tracking, virtual PD effort, end-effector tracking, and arm–torso clearance costs.
6. A rollout that violates the hard arm–torso clearance threshold is removed completely from the MPPI weighted average.

Consequently, this controller is structurally different from a centroidal legged MPC that optimizes contact forces under a prescribed contact-mode schedule. Contact forces are internal MuJoCo variables, and gait timing acts as a kinematic prior rather than a hard contact constraint.

2 Robot Model, State, and Input

2.1 Degrees of Freedom

The MuJoCo model dimensions are

$$n_q = 23, \qquad n_v = 22, \qquad n_u = 16. \tag{1}$$

The actuated coordinates consist of twelve B2 leg joints and the first four Z1 arm joints. Z1 joints 5 and 6 and the gripper are fixed in the reduced model.

The actuator and joint order is

$$\begin{aligned} \boldsymbol{q}_j = [&q_{\mathrm{FR,h}}, q_{\mathrm{FR,t}}, q_{\mathrm{FR,c}}, q_{\mathrm{FL,h}}, q_{\mathrm{FL,t}}, q_{\mathrm{FL,c}}, \\ &q_{\mathrm{RR,h}}, q_{\mathrm{RR,t}}, q_{\mathrm{RR,c}}, q_{\mathrm{RL,h}}, q_{\mathrm{RL,t}}, q_{\mathrm{RL,c}}, q_{a1}, q_{a2}, q_{a3}, q_{a4}]^\top, \end{aligned}$$

where h, t, and c denote hip, thigh, and calf, respectively.

2.2 Observation State

The controller receives the concatenation of MuJoCo `qpos` and `qvel`:

$$\boxed{\boldsymbol{x} = \begin{bmatrix} \boldsymbol{p}_b \\ \boldsymbol{q}_b \\ \boldsymbol{q}_j \\ \boldsymbol{v}_b \\ \boldsymbol{\omega}_b \\ \dot{\boldsymbol{q}}_j \end{bmatrix} \in \mathbb{R}^{45}.} \tag{2}$$

The components are:

- $\boldsymbol{p}_b \in \mathbb{R}^3$: floating-base position in the world frame,
- $\boldsymbol{q}_b = [q_w, q_x, q_y, q_z]^\top \in \mathbb{S}^3$: base quaternion,
- $\boldsymbol{q}_j \in \mathbb{R}^{16}$: actuated leg and arm joint positions,
- $\boldsymbol{v}_b, \boldsymbol{\omega}_b \in \mathbb{R}^3$: base linear and angular velocities,
- $\dot{\boldsymbol{q}}_j \in \mathbb{R}^{16}$: actuated joint velocities.

The exact array layout is summarized below.

Indices	Dimension	Quantity
0:3	3	base position $\boldsymbol{p}_b$
3:7	4	base quaternion $\boldsymbol{q}_b$
7:23	16	actuated joint positions $\boldsymbol{q}_j$
23:26	3	base linear velocity $\boldsymbol{v}_b$
26:29	3	base angular velocity $\boldsymbol{\omega}_b$
29:45	16	actuated joint velocities $\dot{\boldsymbol{q}}_j$

The MuJoCo FULLPHYSICS rollout buffer contains $[t, \boldsymbol{q}_{\text{pos}}^\top, \boldsymbol{q}_{\text{vel}}^\top]^\top \in \mathbb{R}^{46}$. The leading time element is removed before cost evaluation, leaving the 45-dimensional state above.

2.3 Control Input and Actuator Model

The optimized control is the desired position of every actuated joint:

$$\boxed{\boldsymbol{u} = \boldsymbol{q}_{j,\text{des}} \in \mathbb{R}^{16}}, \qquad \boldsymbol{u}_{\min} \leq \boldsymbol{u} \leq \boldsymbol{u}_{\max}. \tag{3}$$

Conceptually, each affine position actuator applies

$$\tau_i = \operatorname{sat}_{[\tau_i^-, \tau_i^+]} \left(k_{p,i}(u_i - q_{j,i}) - k_{d,i}\dot{q}_{j,i} \right). \tag{4}$$

The B2 hip actuators use $(k_p, k_d) = (100, 5)$, the thigh and calf actuators use $(300, 8)$, and all four arm actuators use $(300, 5)$. Joint-position limits and actuator-force limits are read from the MJCF model and applied by action clipping and MuJoCo.

3 Prediction Dynamics and Contact

3.1 Discrete MuJoCo Dynamics

The controller does not implement a closed-form reduced model $\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u})$. One prediction step is instead represented as

$$\boxed{\boldsymbol{x}_{k+1} = F_{\mathrm{MJ}}(\boldsymbol{x}_k, \boldsymbol{u}_k; \boldsymbol{\theta}_{\mathrm{model}}),} \tag{5}$$

where F_{MJ} is evaluated by `rollout.rollout`. The model captures:

- floating-base and articulated rigid-body dynamics,
- position-PD actuators and force saturation,
- gravity, joint damping, and armature,
- foot–ground collision, friction, and MuJoCo soft-contact dynamics.

The default integration interval is $\Delta t = 0.01\,\mathrm{s}$ and the horizon is $H = 40$, giving

$$T_H = H\Delta t = 0.4\,\mathrm{s}. \tag{6}$$

3.2 Consistency Between Prediction and Simulation

The MPPI controller and the closed-loop simulator load separate MuJoCo model instances. During simulator initialization, the contact cone, override flag, `o_solref`, `o_solimp`, friction, and margin are copied from the rollout model to the simulator model. The locomani configuration uses a pyramidal friction cone and

$$\texttt{o\_solref} = [0.02, 1.0]. \tag{7}$$

This synchronization reduces prediction–execution mismatch in contact dynamics.

3.3 Implicit Contact Scheduling

Contact force is not part of $\boldsymbol{u}$, and there are no explicit stance or swing equality constraints. A periodic leg-joint trajectory encourages the intended swing and stance motion, but actual contact is determined by MuJoCo collision detection. The gait scheduler is therefore a phase-aligned kinematic prior, not a strict contact-mode scheduler.

4 Finite-Horizon Stochastic Optimal Control Problem

At each control update, the intended finite-horizon problem can be written as

$$\boxed{\begin{aligned} \min_{\boldsymbol{U}} \quad & J(\boldsymbol{U}; \boldsymbol{x}_0) = \sum_{k=0}^{H-1} \ell_k(\boldsymbol{x}_{k+1}, \boldsymbol{u}_k) + \Phi_{\mathrm{EE}}(\boldsymbol{x}_H), \\ \text{s.t.} \quad & \boldsymbol{x}_{k+1} = F_{\mathrm{MJ}}(\boldsymbol{x}_k, \boldsymbol{u}_k), \\ & \boldsymbol{u}_{\min} \le \boldsymbol{u}_k \le \boldsymbol{u}_{\max}, \\ & d_{k,s} \ge d_{\mathrm{hard}}, \qquad s = 1, 2, 3, \\ & \boldsymbol{x}_0 = \hat{\boldsymbol{x}}(t), \end{aligned}} \tag{8}$$

where $\boldsymbol{U} = [\boldsymbol{u}_0, \ldots, \boldsymbol{u}_{H-1}]$ and $d_{k,s}$ is the signed clearance between arm capsule s and the torso box.

The nonlinear, contact-rich problem in Eq. (8) is not solved with gradients. Instead, $N = 30$ control sequences are sampled around the current nominal trajectory, evaluated through parallel MuJoCo rollouts, and combined by exponential weighting.

5 Nominal Trajectory Construction

5.1 Leg Gait Reference

The default `in_place` gait file has the form

$$\boldsymbol{G} \in \mathbb{R}^{32\times L}, \qquad L = 100, \tag{9}$$

with rows ordered as

$$\boldsymbol{G} = \begin{bmatrix} \boldsymbol{q}_\ell^g \\ \boldsymbol{q}_a^g \\ \dot{\boldsymbol{q}}_\ell^g \\ \dot{\boldsymbol{q}}_a^g \end{bmatrix}, \qquad \boldsymbol{q}_\ell^g, \dot{\boldsymbol{q}}_\ell^g \in \mathbb{R}^{12\times L}. \tag{10}$$

At every controller update, the gait scheduler circularly shifts the column indices by one sample and exposes the next H reference columns.

5.2 Arm IK Reference

For the active world-frame EE target $\boldsymbol{p}_e^*$, the controller solves a position-only damped least-squares IK problem. At iteration r,

$$\boldsymbol{e}^{(r)} = \boldsymbol{p}_e^* - \boldsymbol{p}_e(\boldsymbol{q}_a^{(r)}), \tag{11}$$

$$\Delta\boldsymbol{q}_a^{(r)} = \boldsymbol{J}_a^\top \left(\boldsymbol{J}_a\boldsymbol{J}_a^\top + \lambda_{\mathrm{IK}}^2\boldsymbol{I}\right)^{-1} \boldsymbol{e}^{(r)}, \tag{12}$$

$$\boldsymbol{q}_a^{(r+1)} = \mathrm{clip}\left(\boldsymbol{q}_a^{(r)} + \alpha_{\mathrm{IK}}\Delta\boldsymbol{q}_a^{(r)}\right). \tag{13}$$

The default damping and step size are $\lambda_{\mathrm{IK}} = 0.05$ and $\alpha_{\mathrm{IK}} = 0.7$, with at most 100 iterations.

IK is recomputed from the current posture at every MPPI update. To avoid abrupt changes in the control prior, the new solution is filtered with a smoothing factor of 0.3 and a per-update joint change limit of 0.04 rad. IK only supplies a nominal arm posture; actual EE error is evaluated directly along every rollout.

5.3 Whole-Body Joint Reference and Residual Warm Start

The controller replaces the arm rows of the gait file with the current IK posture:

$$\boldsymbol{r}_{j,k} = \begin{bmatrix} \boldsymbol{q}_{\ell,k}^g \\ \boldsymbol{q}_a^{\mathrm{IK}} \\ \dot{\boldsymbol{q}}_{\ell,k}^g \\ \boldsymbol{0}_4 \end{bmatrix} \in \mathbb{R}^{32}. \tag{14}$$

Its position component defines the nominal action:

$$\bar{\boldsymbol{u}}_k = \begin{bmatrix} \boldsymbol{q}_{\ell,k}^g \\ \boldsymbol{q}_a^{\mathrm{IK}} \end{bmatrix} + \Delta\boldsymbol{u}_k^{\mathrm{warm}}. \tag{15}$$

After optimization, the difference between the selected action and the gait/IK nominal is stored as a residual and shifted by one step. The new horizon tail is initialized with zero residual, so the newly exposed future sample begins at the original gait/IK nominal. During the first 50 controller updates, the nominal is linearly blended from the initial standing control to the gait/IK reference.

6 Sampling and MPPI Update

6.1 Cubic Noise Sampling

The default sampler draws Gaussian perturbations at $K = 4$ temporal knots and interpolates only the perturbation with a cubic spline:

$$\begin{aligned} \boldsymbol{\epsilon}_{n,j} &\sim \mathcal{N}(\mathbf{0}, \mathrm{diag}(\boldsymbol{\sigma}^2)), & j = 1, \ldots, K, \quad (16) \\ \tilde{\boldsymbol{\epsilon}}_{n,0:H-1} &= \mathrm{CubicSpline}(\boldsymbol{\epsilon}_{n,1:K}), & (17) \\ \boldsymbol{u}_{n,k} &= \mathrm{clip}\left(\bar{\boldsymbol{u}}_k + \tilde{\boldsymbol{\epsilon}}_{n,k}, \boldsymbol{u}_{\min}, \boldsymbol{u}_{\max}\right). & (18) \end{aligned}$$

The locomani noise standard deviations are 0.03 for each hip, 0.1 for each thigh and calf, and 0.06 rad for each arm joint. Candidate zero is overwritten with the noise-free nominal, ensuring that every batch contains at least one unperturbed baseline trajectory.

6.2 Cost Normalization and Weights

Let $\mathcal{V}$ be the set of candidates that pass the hard collision test, and let S_n be the total rollout cost. The implementation normalizes costs by the valid-sample range:

$$\begin{aligned} S_{\min} &= \min_{n \in \mathcal{V}} S_n, & (19) \\ \Delta S &= \max_{n \in \mathcal{V}} S_n - S_{\min}, & (20) \\ w_n &= \begin{cases} \exp\left[-\dfrac{1}{\lambda_{\mathrm{MPPI}}} \dfrac{S_n - S_{\min}}{\Delta S}\right], & n \in \mathcal{V},\ \Delta S > 0, \\ 1, & n \in \mathcal{V},\ \Delta S \simeq 0, \\ 0, & n \notin \mathcal{V}. \end{cases} & (21) \end{aligned}$$

The default temperature is $\lambda_{\mathrm{MPPI}} = 0.1$.

The new sequence is the weighted average of the absolute candidate actions:

$$\boxed{\boldsymbol{u}_k^+ = \mathrm{clip}\left(\frac{\sum_{n=1}^N w_n \boldsymbol{u}_{n,k}}{\sum_{n=1}^N w_n + 10^{-10}}\right).} \quad (22)$$

Only $\boldsymbol{u}_0^+$ is executed; the remainder forms the next warm start.

If every candidate violates the hard clearance threshold, no weighted average is formed. The last accepted safe trajectory is shifted by one step, and its final action is repeated at the tail.

7 Cost Function

7.1 State Reference

The horizon state reference is assembled as

$$\boldsymbol{x}_k^{\mathrm{ref}} = \begin{bmatrix} \boldsymbol{p}_b^{\mathrm{ref}} \\ \boldsymbol{q}_b^{\mathrm{ref}} \\ \boldsymbol{q}_{j,k}^{\mathrm{ref}} \\ \boldsymbol{v}_b^{\mathrm{ref}} \\ \boldsymbol{\omega}_b^{\mathrm{ref}} \\ \dot{\boldsymbol{q}}_{j,k}^{\mathrm{ref}} \end{bmatrix}. \quad (23)$$

For locomani, the body pose reference is the `stand` keyframe and the base velocity reference is zero. Before cost evaluation, the current world-frame base linear velocity is transformed into the body-local frame. The joint references come from the gait/IK sequence defined above.

7.2 Body and Joint State Cost

The controller does not penalize a direct four-component quaternion difference. It converts the current and reference base quaternions to roll and pitch and overwrites the orientation slots of the error vector. Conceptually,

$$\boldsymbol{e}_{x,k} = \begin{bmatrix} \boldsymbol{p}_{b,k} - \boldsymbol{p}_b^{\text{ref}} \\ e_{\text{roll},k} \\ e_{\text{pitch},k} \\ 0 \\ 0 \\ \boldsymbol{q}_{j,k} - \boldsymbol{q}_{j,k}^{\text{ref}} \\ \boldsymbol{v}_{b,k} - \boldsymbol{v}_b^{\text{ref}} \\ \boldsymbol{\omega}_{b,k} - \boldsymbol{\omega}_b^{\text{ref}} \\ \dot{\boldsymbol{q}}_{j,k} - \dot{\boldsymbol{q}}_{j,k}^{\text{ref}} \end{bmatrix}, \tag{24}$$

and

$$\ell_{x,k} = \boldsymbol{e}_{x,k}^\top \boldsymbol{Q} \boldsymbol{e}_{x,k}. \tag{25}$$

The main diagonal state weights are:

- base height: 1000,
- base roll and pitch: 5000 each,
- all 16 joint positions: 1200 each,
- each base linear and angular velocity component: 0.1.

The current weights for base x/y, yaw, and joint-velocity state errors are zero.

7.3 Virtual PD Effort Cost

The control penalty is applied to a virtual PD effort rather than directly to the desired joint positions:

$$\tilde{\boldsymbol{\tau}}_k = \boldsymbol{K}_p(\boldsymbol{u}_k - \boldsymbol{q}_{j,k}) - \boldsymbol{K}_d(\dot{\boldsymbol{q}}_{j,k} - \dot{\boldsymbol{q}}_{j,k}^{\text{ref}}), \tag{26}$$

$$\ell_{u,k} = \tilde{\boldsymbol{\tau}}_k^\top \boldsymbol{R} \tilde{\boldsymbol{\tau}}_k, \qquad \boldsymbol{R} = 10^{-3}\boldsymbol{I}_{16}. \tag{27}$$

This is a cost-side virtual effort. It is not exactly the actuator force applied by MuJoCo because the expression does not include force saturation.

7.4 End-Effector Pose Cost

Let $\boldsymbol{p}_{e,k}$ and $\boldsymbol{q}_{e,k}$ be the EE position and quaternion obtained from MuJoCo FK or rollout sensors. The position cost is

$$\ell_{p,k} = w_p \left\| \boldsymbol{p}_{e,k} - \boldsymbol{p}_e^* \right\|^2, \qquad w_p = 5000. \tag{28}$$

The quaternion double cover is handled using the absolute quaternion dot product:

$$e_{R,k} = 2 \arccos \left[\text{clip} \left(\left| \boldsymbol{q}_{e,k}^\top \boldsymbol{q}_e^* \right|, -1, 1 \right) \right], \tag{29}$$

$$\ell_{R,k} = w_R e_{R,k}^2. \tag{30}$$

The current configuration sets $w_R = 0$, so EE orientation tracking is disabled.

An additional terminal multiplier $\gamma_T = 10$ is applied to the final EE cost:

$$\Phi_{\text{EE}}(\boldsymbol{x}_H) = \gamma_T(\ell_{p,H-1} + \ell_{R,H-1}). \tag{31}$$

Because the same error is already present in the final stage cost, its total EE coefficient at the final step is $1 + \gamma_T = 11$.

7.5 Total Stage and Rollout Cost

With the collision soft cost $\ell_{c,k}$ defined in the next section,

$$\ell_k = \ell_{x,k} + \ell_{u,k} + \ell_{p,k} + \ell_{R,k} + \ell_{c,k}, \tag{32}$$

and the complete candidate cost is

$$\boxed{S_n = \sum_{k=0}^{H-1} \ell_k^{(n)} + \gamma_T \left(\ell_{p,H-1}^{(n)} + \ell_{R,H-1}^{(n)}\right).} \tag{33}$$

8 Arm–Torso Collision Model

8.1 Geometry

Four frames—shoulder, elbow, wrist, and EE—define three arm capsules. Their radii are

$$\boldsymbol{r}_{\text{cap}} = [0.030, 0.030, 0.0375]\ \text{m}. \tag{34}$$

The torso is approximated by the oriented box associated with the `base_collision` geometry, whose half-size is

$$\boldsymbol{h}_b = [0.25, 0.14, 0.075]\ \text{m}. \tag{35}$$

The first 0.07 m of the proximal capsule is excluded to prevent the structural shoulder mount from producing a false collision.

For capsule s, the signed clearance $d_{k,s}$ is the distance between the capsule centerline and the torso box minus the capsule radius. Thus $d > 0$ means separation, $d = 0$ means contact, and $d < 0$ means penetration.

8.2 Soft Cost and Hard Rejection

Inside the safe distance $d_{\text{safe}} = 0.10\,\text{m}$, the controller uses the quadratic penalty

$$\boxed{\ell_{c,k} = w_c \sum_{s=1}^{3} \left[\max(d_{\text{safe}} - d_{k,s}, 0)\right]^2, \qquad w_c = 10^4.} \tag{36}$$

A candidate is also marked invalid if any state in its horizon violates

$$d_{k,s} \geq d_{\text{hard}} = 0.005\,\text{m}. \tag{37}$$

Invalid candidates receive zero MPPI weight.

For efficiency, each arm segment is first sampled at nine points. A Lipschitz lower bound identifies segments that could enter the soft-cost region, and only those segments receive an exact segment–axis-aligned-box distance calculation in the box frame.

9 End-Effector Waypoints and Task Progression

The locomani task tracks a sequence of world-frame EE positions:

$$\begin{aligned}
\boldsymbol{p}_{e,0}^* &= [0.85, 0.00, 0.80]^\top, \\
\boldsymbol{p}_{e,1}^* &= [2.00, 0.15, 0.80]^\top, \\
\boldsymbol{p}_{e,2}^* &= [0.85, 0.15, 0.48]^\top, \\
\boldsymbol{p}_{e,3}^* &= [0.00, 0.00, 0.48]^\top, \\
\boldsymbol{p}_{e,4}^* &= [-1.00, 0.00, 0.80]^\top.
\end{aligned} \tag{38}$$

Some targets are outside the fixed-base reach of the arm. Their world-frame EE cost can therefore induce base locomotion in addition to arm motion.

The goal check is

$$\|\boldsymbol{p}_e - \boldsymbol{p}_e^*\| \leq 0.03\,\mathrm{m}, \qquad 1 - \left|\boldsymbol{q}_e^\top \boldsymbol{q}_e^*\right| \leq 1.0. \tag{39}$$

Since the absolute quaternion dot product lies in $[0, 1]$, the configured orientation threshold is always satisfied. In practice, waypoint completion depends only on EE position. The waypoint timer advances only when the simulator calls `next_goal()` after a successful goal check.

10 Complete Receding-Horizon Algorithm

Algorithm 1 `legged_mani` locomani MPPI update

Require: Current observation $\hat{\boldsymbol{x}}$, gait phase, previous safe plan

Update arm IK for the active EE waypoint.
Combine the leg gait and arm IK to refresh nominal plan $\bar{\boldsymbol{U}}$.
Sample Gaussian noise at four knots and apply cubic interpolation to generate $N = 30$ candidate action sequences.
Replace candidate zero with the noise-free nominal sequence.
Roll out all candidates for $H = 40$ steps in MuJoCo.
Evaluate state, effort, EE, and collision costs S_n.
Assign zero weight to candidates that violate hard clearance.
if at least one valid candidate exists **then**
    Normalize valid costs and compute exponential weights.
    Compute $\boldsymbol{U}^+$ as the weighted mean of absolute candidate actions.
    Store $\boldsymbol{U}^+$ as the most recent safe plan.
else
    Shift and reuse the previous safe plan as $\boldsymbol{U}^+$.
end if
Shift the optimized gait/IK residual and advance the gait phase.
return First desired joint-position command $\boldsymbol{u}_0^+$

The default closed-loop simulator invokes this update at 100 Hz and applies the first action for one MuJoCo step. It then evaluates the goal condition and updates the waypoint manager if the active target was reached.

11 Default Configuration Summary

Parameter	Default value
Robot	Unitree B2 with reduced four-DoF Z1 arm
State / input dimension	45 / 16
Input interpretation	desired joint position
Time step / horizon	0.01 s / 40 steps (0.4 s)
Samples / workers	30 / 5
Temperature	0.1
Sampler	cubic perturbation, four knots
Initial keyframe	`stand`
Default gait	`in_place`, 100 samples per cycle
Rollout contact	pyramidal cone, `o_solref`$= [0.02, 1.0]$
EE position/orientation weights	5000 / 0
EE terminal scale	10
Collision safe/hard distances	0.10/0.005 m
Collision soft weight	10^4

12 Comparison with OCS2 Legged MPC

Aspect	OCS2 legged MPC	`legged_mani` locomani
Dynamics	Centroidal/SRBD or full centroidal	MuJoCo full multibody rollout
Optimized input	Contact forces and joint velocities	Desired joint positions
Contact transitions	Prescribed mode schedule	Gait prior and MuJoCo collision
Constraint handling	Equalities, inequalities, barriers	Action clipping and rollout rejection
Optimization method	SQP, DDP, or IPM	Sampling-based MPPI-style update
Derivative information	Dynamics and cost derivatives	Derivative-free rollout scoring
Manipulation objective	Not part of the basic legged example	EE world-pose cost and arm IK prior
Collision avoidance	Added as problem-specific constraints	Three arm capsules against torso OBB

13 Implementation Caveats and Limitations

1. **Difference from canonical MPPI.** The implementation normalizes cost by the batch min–max range and averages absolute candidate actions instead of applying a weighted noise increment with a control-noise correction term. Consequently, $\lambda = 0.1$ controls selection pressure over normalized relative cost, not temperature in the original physical cost units.

2. **EE orientation is disabled.** The quaternion cost is zero and the orientation success threshold is nonrestrictive. The current task is an EE position task. Enforcing a general six-dimensional pose with a four-DoF arm would require a reachable-task analysis and different objective design.

3. **The collision model is incomplete.** Hard rejection covers only three arm capsules against one torso box. It does not guarantee arm–leg, arm–ground, gripper–environment, or general

self-collision safety.

4. **Gait timing is not a contact constraint.** If gait tracking becomes weak, the desired stance/swing timing is not guaranteed. If joint tracking is too strong, it can suppress the whole-body motion needed to reduce EE error.

5. **Cost-side effort differs from applied actuator force.** The virtual PD effort omits actuator-force saturation. Large tracking errors can therefore produce a discrepancy between the effort penalty and the force actually applied in MuJoCo.

6. **Real-time performance must be measured.** Every update evaluates 30×40 contact-dynamics steps and rollout sensors. Sustained 100 Hz operation depends on the target CPU and should be verified from wall-clock timing rather than the nominal simulation time step.

7. **Waypoint dwell is not necessarily continuous.** The waypoint timer increments only on successful goal checks and is not reset when the EE leaves the threshold. It therefore counts successful invocations rather than enforcing a strict continuous dwell interval.

14 Recommended Validation

The following measurements are useful for connecting the formulation to closed-loop performance:

1. mean, 95th, and 99th percentile update time and the 10 ms deadline-miss rate,
2. fraction of valid rollouts, minimum torso clearance, and the number of all-invalid fallbacks,
3. EE position error and waypoint transition time,
4. base height, roll/pitch, foot contacts, and slip,
5. gait/IK nominal, selected actions, and warm-start residual magnitude,
6. actuator saturation rate and virtual PD effort,
7. one-step error between the predicted rollout and the executed simulator state.

Useful ablations include: soft collision cost without hard rejection, the combined soft/hard scheme, zero arm exploration noise, removal of the terminal EE multiplier, and removal of gait-residual warm starting.

15 Pipeline Summary

Current MuJoCo state $\hat{\boldsymbol{x}}_t$
$\downarrow$
Phase-aligned leg gait + arm IK for the active EE target
$\downarrow$
Whole-body nominal action + warm-started residual
$\downarrow$
Cubic Gaussian sampling ($30 \times 40 \times 16$)
$\downarrow$
Parallel MuJoCo rollout + EE/FK sensors
$\downarrow$
State/effort + EE + collision cost
$\downarrow$
Hard-collision rejection + exponential weighting
$\downarrow$
Execute the first desired joint-position action
$\downarrow$
Shift gait phase and residual for the next MPC update

The defining design choice is to sample desired joint-position trajectories through full MuJoCo dynamics rather than optimizing contact forces over a reduced-order model. The gait and IK modules provide a structured prior, while the MPPI update searches for whole-body residuals that balance stability, EE tracking, virtual effort, and torso clearance.